Almost-everywhere Fourier convergence for continuous periodic functions

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Allowed standard prerequisites (editorial).

- Standard measure theory, dominated/monotone convergence, Fourier Hilbert-space identities and elementary doubling-space facts are cited as in the author Blueprint.
- For the classical theorem the metric-space Carleson theorem is an explicitly stated named prerequisite, with its separate complete proof retained in the companion material.

0.1 Proof of Cotlar's Inequality

1 Proof of Metric Space Carleson, overview

With Euler's formula to represent pure frequencies in mind, a trigonometric polynomial can be expressed as

$$S_N(x) := \sum_{n=-N}^{N} c_n e^{inx} \ . \tag{1.0.1}$$

The Fourier series is then defined as the limit f of such a sequence S_N as N tends to ∞. Fourier's vision to represent rather general functions raises two fundamental questions. The first question is to identify the appropriate choice of coefficients c_n to use to represent a given f. The second question addresses the convergence of S_N to f.

The first question has a fairly canonical and standard answer, provided by the Fourier integral formula:

$$c_n := \hat{f}_n := \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-inx} \, dx, \tag{1.0.2}$$

1.1 Statement of the metric space Carleson theorems

Let

$$a \geq 4 \tag{1.1.1}$$

be a natural number that as it gets larger will allow for more general applications of Theorem 1.1.1 but will worsen the constants in the estimates.

A doubling metric measure space (X, ρ, μ, a) is a complete and locally compact metric space (X, ρ) equipped with a non-zero locally finite Borel measure μ that satisfies the doubling condition that for all $x \in X$ and all $R > 0$ we have

$$\mu(B(x, 2R)) \leq 2^a \mu(B(x, R)) \,, \tag{1.1.2}$$

where we have denoted by $B(x, R)$ the open ball of radius R centered at x:

$$B(x, R) := \{y \in X : \rho(x, y) < R\}. \tag{1.1.3}$$

In a doubling metric measure space the closed balls are compact and μ is positive on all non-empty open sets.

A collection Θ of real valued continuous functions on the doubling metric measure space (X, ρ, μ, a) is called compatible, if there is a point $o \in X$ where all the functions are equal to 0, and if there exists for each ball $B \subset X$ a metric d_B on Θ, such that the following five properties (1.1.4), (1.1.5), (1.1.6), (1.1.7), and (1.1.8) are satisfied. For every ball $B \subset X$

$$\sup_{x,y \in B} |\vartheta(x) - \vartheta(y) - \theta(x) + \theta(y)| \leq d_B(\vartheta, \theta) \,. \tag{1.1.4}$$

For any two balls $B_1 = B(x_1, R)$, $B_2 = B(x_2, 2R)$ in X with $x_1 \in B_2$ and any $\vartheta, \theta \in \Theta$,

$$d_{B_2}(\vartheta, \theta) \leq 2^a d_{B_1}(\vartheta, \theta). \tag{1.1.5}$$

For any two balls B_1, B_2 in X with $B_1 \subset B_2$ and any $\vartheta, \theta \in \Theta$

$$d_{B_1}(\vartheta, \theta) \leq d_{B_2}(\vartheta, \theta) \tag{1.1.6}$$

and for any two balls $B_1 = B(x_1, R)$, $B_2 = B(x_2, 2^a R)$ with $B_1 \subset B_2$, and $\vartheta, \theta \in \Theta$,

$$2 d_{B_1}(\vartheta, \theta) \leq d_{B_2}(\vartheta, \theta). \tag{1.1.7}$$

For every ball B in X and every d_B-ball $\tilde{B}$ of radius $2R$ in Θ, there is a collection $\mathcal{B}$ of at most 2^a many d_B-balls of radius R covering $\tilde{B}$, that is,

$$\tilde{B} \subset \bigcup \mathcal{B}. \tag{1.1.8}$$

Further, a compatible collection Θ is called cancellative, if for any ball B in X of radius R, any Lipschitz function $\varphi : X \to \mathbb{C}$ supported on B, and any $\vartheta, \theta \in \Theta$ we have

$$\Big| \int_B e(\vartheta(x) - \theta(x)) \varphi(x) d\mu(x) \Big| \leq 2^a \mu(B) \|\varphi\|_{\mathrm{Lip}(B)} (1 + d_B(\vartheta, \theta))^{-\frac{1}{a}}, \tag{1.1.9}$$

where $\| \cdot \|_{\mathrm{Lip}(B)}$ denotes the inhomogeneous Lipschitz norm on B:

$$\|\varphi\|_{\mathrm{Lip}(B)} = \sup_{x \in B} |\varphi(x)| + R \sup_{x,y \in B, x \neq y} \frac{|\varphi(x) - \varphi(y)|}{\rho(x, y)} \,.$$

A one-sided Calderón–Zygmund kernel K on the doubling metric measure space (X, ρ, μ, a) is a measurable function

$$K : X \times X \to \mathbb{C} \tag{1.1.10}$$

such that for all $x, y', y \in X$ with $x \neq y$, we have

$$|K(x,y)| \leq \frac{2^{a^3}}{V(x,y)} \tag{1.1.11}$$

and if $2\rho(y,y') \leq \rho(x,y)$, then

$$|K(x,y) - K(x,y')| \leq \left(\frac{\rho(y,y')}{\rho(x,y)}\right)^{\frac{1}{a}} \frac{2^{a^3}}{V(x,y)}, \tag{1.1.12}$$

where

$$V(x,y) := \mu(B(x, \rho(x,y))).$$

Define the maximally truncated non-tangential singular integral T_* associated with K by

$$T_* f(x) := \sup_{R_1 < R_2} \sup_{\rho(x,x') < R_1} \left| \int_{R_1 < \rho(x',y) < R_2} K(x',y) f(y) \, \mathrm{d}\mu(y) \right|. \tag{1.1.13}$$

We define the generalized Carleson operator T by

$$Tf(x) := \sup_{\vartheta \in \Theta} \sup_{0 < R_1 < R_2} \left| \int_{R_1 < \rho(x,y) < R_2} K(x,y) f(y) e(\vartheta(y)) \, \mathrm{d}\mu(y) \right|, \tag{1.1.14}$$

where $e(r) = e^{ir}$.

Our first main result is the following restricted weak type estimate for T in the range $1 < q \leq 2$, which by interpolation techniques recovers L^q estimates for the open range $1 < q < 2$.

Theorem 1.1.1 (metric space Carleson)**.** *For all integers $a \geq 4$ and real numbers $1 < q \leq 2$ the following holds. Let (X, ρ, μ, a) be a doubling metric measure space. Let Θ be a cancellative compatible collection of functions and let K be a one-sided Calderón–Zygmund kernel on (X, ρ, μ, a). Assume that for every bounded measurable function g on X supported on a set of finite measure we have*

$$\|T_* g\|_2 \leq 2^{a^3} \|g\|_2 , \tag{1.1.15}$$

where T_* is defined in (1.1.13)*. Then for all Borel sets F and G in X and all Borel functions $f : X \to \mathbb{C}$ with $|f| \leq \mathbf{1}_F$, we have, with T defined in* (1.1.14)*,*

$$\left| \int_G Tf \, \mathrm{d}\mu \right| \leq \frac{2^{443a^3}}{(q-1)^6} \mu(G)^{1-\frac{1}{q}} \mu(F)^{\frac{1}{q}} . \tag{1.1.16}$$

In some applications, such as the Walsh-case of Carleson's theorem, the kernel K naturally depends also on the modulation functions ϑ. The fact that we don't assume Hölder-continuity of the kernel K in the first argument allows us to also capture this situation.

We now state another Theorem with a slightly weaker replacement of the assumption (1.1.15). It implies the Walsh-case of Carleson's theorem. For a Borel function $Q : X \to \Theta$, and $\vartheta \in \Theta$ and $x \in X$ define

$$R_Q(\vartheta, x) = \sup\{r > 0 : d_{B(x,r)}(\vartheta, Q(x)) < 1\} \tag{1.1.17}$$

and define further

$$T_Q^{\vartheta} f(x) := \sup_{R_1<R_2} \sup_{\rho(x,x')<R_1} \left| \int_{R_1<\rho(x',y)<\min\{R_2,R_Q(\vartheta,x')\}} K(x',y) f(y)\, \mathrm{d}\mu(y) \right| \tag{1.1.18}$$

Define further the linearized generalized Carleson operator T_Q by

$$T_Q f(x) := \sup_{0<R_1<R_2} \left| \int_{R_1<\rho(x,y)<R_2} K(x,y) f(y) e(Q(x)(y))\, \mathrm{d}\mu(y) \right| , \tag{1.1.19}$$

where again $e(r) = e^{ir}$.

Theorem 1.1.2 (linearised metric Carleson)**.** *For all integers $a \geq 4$ and real numbers $1 < q \leq 2$ the following holds. Let (X, ρ, μ, a) be a doubling metric measure space. Let Θ be a cancellative compatible collection of functions. Let $Q : X \to \Theta$ be a Borel function with finite range. Let K be a one-sided Calderón–Zygmund kernel on (X, ρ, μ, a). Assume that for every $\vartheta \in \Theta$ and every bounded measurable function g on X supported on a set of finite measure we have*

$$\|T_Q^{\vartheta} g\|_2 \leq 2^{a^3} \|g\|_2 , \tag{1.1.20}$$

where T_Q^{ϑ} is defined in (1.1.18)*. Then for all Borel sets F and G in X and all Borel functions $f : X \to \mathbb{C}$ with $|f| \leq \mathbf{1}_F$, we have, with T_Q defined in* (1.1.19)*,*

$$\left| \int_G T_Q f \, \mathrm{d}\mu \right| \leq \frac{2^{443a^3}}{(q-1)^6} \mu(G)^{1-\frac{1}{q}} \mu(F)^{\frac{1}{q}} . \tag{1.1.21}$$

2 Two-sided Metric Space Carleson

We prove a variant of Theorem 1.1.1 for a two-sided Calderón–Zygmund kernel on the doubling metric measure space (X, ρ, μ, a), i.e. a one-sided Calderón–Zygmund kernel K which additionally satisfies for all $x, x', y \in X$ with $x \neq y$ and $2\rho(x, x') \leq \rho(x, y)$,

$$|K(x,y) - K(x',y)| \leq \left(\frac{\rho(x,x')}{\rho(x,y)} \right)^{\frac{1}{a}} \frac{2^{a^3}}{V(x,y)} . \tag{2.0.1}$$

By the additional regularity, we can weaken the assumption (1.1.15) to a family of operators that is easier to work with in applications. Namely, for $r > 0$, $x \in X$, and a bounded, measurable function $f : X \to \mathbb{C}$ supported on a set of finite measure, we define

$$T_r f(x) := \int_{r \leq \rho(x,y)} K(x,y) f(y)\, d\mu(y) = \int_{X \setminus B(x,r)} K(x,y) f(y)\, d\mu(y). \tag{2.0.2}$$

Theorem 2.0.1 (two-sided metric space Carleson)**.** *For all integers $a \geq 4$ and real numbers $1 < q \leq 2$ the following holds. Let (X, ρ, μ, a) be a doubling metric measure space. Let Θ be a cancellative compatible collection of functions and let K be a two-sided Calderón–Zygmund kernel on (X, ρ, μ, a). Assume that for every bounded measurable function g on X supported on a set of finite measure and all $r > 0$ we have*

$$\|T_r g\|_2 \leq 2^{a^3} \|g\|_2 . \tag{2.0.3}$$

Then for all Borel sets F and G in X and all Borel functions $f : X \to \mathbb{C}$ with $|f| \leq \mathbf{1}_F$, we have, with T defined in (1.1.14)*,*

$$\left| \int_G T f \, \mathrm{d}\mu \right| \leq \frac{2^{474a^3}}{(q-1)^6} \mu(G)^{1-\frac{1}{q}} \mu(F)^{\frac{1}{q}} . \tag{2.0.4}$$

For the remainder of this chapter, fix an integer $a \geq 4$, a doubling metric measure space (X, ρ, μ, a) and a two-sided Calderón–Zygmund kernel K as in Theorem 2.0.1. The following lemma is proved in Section 0.1.

Lemma 2.0.2 (nontangential-from-simple)**.** *Assume* (2.0.3) *holds. Then, for every bounded measurable function $g : X \to \mathbb{C}$ supported on a set of finite measure we have*

$$\|T_* g\|_2 \leq 2^{3a^3} \|g\|_2. \tag{2.0.5}$$

3 Proof of The Classical Carleson Theorem

The convergence of partial Fourier sums is proved in Section 3.1 in two steps. In the first step, we establish convergence on a suitable dense subclass of functions. We choose smooth functions as subclass, the convergence is stated in Theorem 3.1.2 and proved in Section 3.2. In the second step, one controls the relevant error of approximating a general function by a function in the subclass. This is stated in Theorem 3.1.3 and proved in Section 3.6. The proof relies on a bound on the real Carleson maximal operator stated in Theorem 3.1.5 and proved in Section 3.7, which involves showing that the real line fits into the setting of Section 1. This latter proof refers to the two-sided variant of the Carleson Theorem 2.0.1. Two assumptions in Theorem 1.1.1 require more work. The boundedness of the operator T_r defined in (2.0.2) is established in 3.1.6. This lemma is proved in Section 3.3. The cancellative property is verified by Theorem 3.1.7, which is proved in Section 3.4. Several further auxiliary lemmas are stated and proved in Section 3.1, the proof of one of these auxiliary lemmas, Theorem 3.1.10, is done in Section 3.5.

All subsections past Section 3.1 are mutually independent.

3.1 The classical Carleson theorem

Let a uniformly continuous 2π-periodic function $f : \mathbb{R} \to \mathbb{C}$ and $\epsilon > 0$ be given. Let

$$C_{a,q} := \frac{2^{474a^3}}{(q-1)^6} \tag{3.1.1}$$

denote the constant from Theorem 2.0.1. Define

$$\epsilon' := \frac{\epsilon}{4C_\epsilon}\,, \tag{3.1.2}$$

where

$$C_\epsilon = \left(\frac{8}{\pi\epsilon}\right)^{\frac{1}{2}} C_{4,2} + \pi\,.$$

Since f is continuous and periodic, f is uniformly continuous. Thus, there is a $0 < \delta < \pi$ such that for all $x, x' \in \mathbb{R}$ with $|x - x'| \leq \delta$ we have

$$|f(x) - f(x')| \leq \epsilon'\,. \tag{3.1.3}$$

Define

$$f_0 := f * \phi_\delta, \tag{3.1.4}$$

where ϕ_δ is a nonnegative smooth bump function with $\operatorname{supp}(\phi_\delta) \subset (-\delta, \delta)$ and $\int_{\mathbb{R}} \phi_\delta(x)\,dx = 1$.

Lemma 3.1.1 (smooth approximation)**.** *The function f_0 is 2π-periodic. The function f_0 is smooth (and therefore measurable). The function f_0 satisfies for all $x \in \mathbb{R}$:*

$$|f(x) - f_0(x)| \leq \epsilon'\,, \tag{3.1.5}$$

Proof. Periodicity follows directly from the definitions. The other properties are part of the Lean library. □

We prove in Section 3.2:

Lemma 3.1.2 (convergence for smooth)**.** *There exists some $N_0 \in \mathbb{N}$ such that for all $N > N_0$ and $x \in [0, 2\pi]$ we have*

$$|S_N f_0(x) - f_0(x)| \leq \frac{\epsilon}{4}\,. \tag{3.1.6}$$

We prove in Section 3.6:

Lemma 3.1.3 (control approximation effect)**.** *There is a set $E \subset \mathbb{R}$ with Lebesgue measure $|E| \leq \epsilon$ such that for all*

$$x \in [0, 2\pi) \setminus E \tag{3.1.7}$$

we have

$$\sup_{N \geq 0} |S_N f(x) - S_N f_0(x)| \leq \frac{\epsilon}{4}\,. \tag{3.1.8}$$

We are now ready to prove Theorem 3.7.13. We first prove a version with explicit exceptional sets.

Theorem 3.1.4 (classical Carleson with exceptional sets)**.** *Let f be a 2π-periodic complex-valued continuous function on $\mathbb{R}$. For all $\epsilon > 0$, there exists a Borel set $E \subset [0, 2\pi]$ with Lebesgue measure $|E| \leq \epsilon$ and a positive integer N_0 such that for all $x \in [0, 2\pi] \setminus E$ and all integers $N > N_0$, we have*

$$|f(x) - S_N f(x)| \leq \epsilon. \tag{3.1.9}$$

Proof. Let N_0 be as in Theorem 3.1.2. For every

$$x \in [0, 2\pi) \setminus E\,, \tag{3.1.10}$$

and every $N > N_0$ we have by the triangle inequality

$$\begin{aligned} &|f(x) - S_N f(x)| \\ &\leq |f(x) - f_0(x)| + |f_0(x) - S_N f_0(x)| + |S_N f_0(x) - S_N f(x)|\,. \end{aligned} \tag{3.1.11}$$

Using Theorems 3.1.1 to 3.1.3, we estimate (3.1.11) by

$$\leq \epsilon' + \frac{\epsilon}{4} + \frac{\epsilon}{4} \leq \epsilon\,. \tag{3.1.12}$$

This shows (3.1.9) for the given E and N_0. □

Now we turn to the proof of the statement of Carleson theorem given in the introduction.

Let $\kappa : \mathbb{R} \to \mathbb{C}$ be the function defined by $\kappa(0) = 0$ and for $0 < |x| < 1$

$$\kappa(x) = \frac{1 - |x|}{1 - e^{ix}} \tag{3.1.13}$$

and for $|x| \geq 1$,

$$\kappa(x) = 0\,. \tag{3.1.14}$$

Note that this function is continuous at every point x with $|x| > 0$.

The proof of Theorem 3.1.3 will use the following Theorem 3.1.5, which itself is proven in Section 3.7 as an application of Theorem 1.1.1.

Lemma 3.1.5 (real Carleson). *Let F, G be Borel subsets of $\mathbb{R}$ with finite measure. Let f be a bounded measurable function on $\mathbb{R}$ with $|f| \le \mathbf{1}_F$. Then*

$$\left| \int_G Tf(x)\,dx \right| \le C_{4,2} |F|^{\frac{1}{2}} |G|^{\frac{1}{2}}\,, \tag{3.1.15}$$

where

$$Tf(x) = \sup_{n\in\mathbb{Z}} \sup_{r>0} \left| \int_{r<|x-y|<1} f(y)\kappa(x-y)e^{iny}\,dy \right|. \tag{3.1.16}$$

One of the main assumptions of Theorem 2.0.1, concerning the operator T_r defined in (2.0.2), is verified by the following lemma, which is proved in Section 3.3.

Lemma 3.1.6 (Hilbert strong 2 2). *Let $0 < r$. Let f be a bounded, measurable function on $\mathbb{R}$. Then*

$$\|H_r f\|_2 \le 2^9 \|f\|_2, \tag{3.1.17}$$

where

$$H_r f(x) := T_r f(x) = \int_{r\le|x-y|} \kappa(x-y) f(y)\,dy \tag{3.1.18}$$

The next lemma will be used to verify that the collection Θ of modulation functions in our application of Theorem 1.1.1 satisfies the condition (1.1.9). It is proved in Section 3.4.

Lemma 3.1.7 (van der Corput). *Let $\alpha \le \beta$ be real numbers. Let $g : \mathbb{R} \to \mathbb{C}$ be a measurable function and assume*

$$\|g\|_{Lip(\alpha,\beta)} := \sup_{\alpha\le x\le\beta} |g(x)| + \frac{|\beta-\alpha|}{2} \sup_{\alpha\le x<y\le\beta} \frac{|g(y)-g(x)|}{|y-x|} < \infty\,. \tag{3.1.19}$$

Then for any $\alpha \le \beta$ and $n \in \mathbb{Z}$ we have

$$\int_\alpha^\beta g(x)e^{inx}\,dx \le 2\pi|\beta-\alpha|\|g\|_{Lip(\alpha,\beta)}(1+|n||\beta-\alpha|)^{-1}\,. \tag{3.1.20}$$

We close this section with five lemmas that are used across the following subsections.

Lemma 3.1.8 (Dirichlet kernel). *We have for every 2π-periodic bounded measurable f and every $N \ge 0$*

$$S_N f(x) = \frac{1}{2\pi} \int_0^{2\pi} f(y) K_N(x-y)\,dy \tag{3.1.21}$$

where K_N is the 2π-periodic continuous function of $\mathbb{R}$ given by

$$\sum_{n=-N}^{N} e^{inx'}\,. \tag{3.1.22}$$

We have for $e^{ix'} \ne 1$ that

$$K_N(x') = \frac{e^{iNx'}}{1-e^{-ix'}} + \frac{e^{-iNx'}}{1-e^{ix'}}\,. \tag{3.1.23}$$

Proof. We have by definitions and interchanging sum and integral

$$\begin{aligned} S_N f(x) &= \sum_{n=-N}^{N} \hat{f}_n e^{inx} \\ &= \sum_{n=-N}^{N} \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{in(x-y)}\, dy \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(y) \sum_{n=-N}^{N} e^{in(x-y)}\, dy\,. \end{aligned} \tag{3.1.24}$$

This proves the first statement of the lemma. By a telescoping sum, we have for every $x' \in \mathbb{R}$

$$\left(e^{\frac{1}{2}ix'} - e^{-\frac{1}{2}ix'}\right) \sum_{n=-N}^{N} e^{inx'} = e^{(N+\frac{1}{2})ix'} - e^{-(N+\frac{1}{2})ix'}\,. \tag{3.1.25}$$

If $e^{ix'} \neq 1$, the first factor on the left-hand side is not 0 and we may divide by this factor to obtain

$$\sum_{n=-N}^{N} e^{inx'} = \frac{e^{i(N+\frac{1}{2})x'}}{e^{\frac{1}{2}ix'} - e^{-\frac{1}{2}ix'}} - \frac{e^{-i(N+\frac{1}{2})x'}}{e^{\frac{1}{2}ix'} - e^{-\frac{1}{2}ix'}} = \frac{e^{iNx'}}{1 - e^{-ix'}} + \frac{e^{-iNx'}}{1 - e^{ix'}}\,. \tag{3.1.26}$$

This proves the second part of the lemma. □

Lemma 3.1.9 (lower secant bound)**.** *Let $\eta > 0$ and $-2\pi + \eta \leq x \leq 2\pi - \eta$ with $|x| \geq \eta$. Then*

$$|1 - e^{ix}| \geq \frac{2}{\pi}\eta \tag{3.1.27}$$

Proof. We have

$$|1 - e^{ix}| = \sqrt{(1 - \cos(x))^2 + \sin^2(x)} \geq |\sin(x)|\,.$$

If $0 \leq x \leq \frac{\pi}{2}$, then we have from concavity of sin on $[0, \pi]$ and $\sin(0) = 0$ and $\sin(\frac{\pi}{2}) = 1$

$$|\sin(x)| \geq \frac{2}{\pi} x \geq \frac{2}{\pi}\eta\,.$$

When $x \in \frac{m\pi}{2} + [0, \frac{\pi}{2}]$ for $m \in \{-4, -3, -2, -1, 1, 2, 3\}$ one can argue similarly. □

The following lemma will be proved in Section 3.5.

Lemma 3.1.10 (spectral projection bound)**.** *Let f be a bounded 2π-periodic measurable function. Then, for all $N \geq 0$*

$$\|S_N f\|_{L^2[0,2\pi]} \leq \|f\|_{L^2[0,2\pi]}. \tag{3.1.28}$$

Lemma 3.1.11 (Hilbert kernel bound)**.** *For $x, y \in \mathbb{R}$ with $x \neq y$ we have*

$$|\kappa(x-y)| \leq 2^2 (2|x-y|)^{-1} \,. \tag{3.1.29}$$

Proof. Fix $x \neq y$. If $\kappa(x-y)$ is zero, then (3.1.29) is evident. Assume $\kappa(x-y)$ is not zero, then $0 < |x-y| < 1$. We have

$$|\kappa(x-y)| = \left| \frac{1-|x-y|}{1-e^{i(x-y)}} \right| . \tag{3.1.30}$$

We estimate with Theorem 3.1.9

$$|\kappa(x-y)| \leq \frac{1}{|1-e^{i(x-y)}|} \leq \frac{2}{|x-y|} \,. \tag{3.1.31}$$

This proves (3.1.29) in the given case and completes the proof of the lemma. □

Lemma 3.1.12 (Hilbert kernel regularity)**.** *For $x, y, y' \in \mathbb{R}$ with $x \neq y, y'$ and*

$$2|y-y'| \leq |x-y| \,, \tag{3.1.32}$$

we have

$$|\kappa(x-y) - \kappa(x-y')| \leq 2^8 \frac{1}{|x-y|} \frac{|y-y'|}{|x-y|} \,. \tag{3.1.33}$$

Proof. Upon replacing y by $y-x$ and y' by $y'-x$ on the left-hand side of (3.1.32), we can assume that $x = 0$. Then the assumption (3.1.32) implies that y and y' have the same sign. Since $\kappa(y) = \bar{\kappa}(-y)$ we can assume that they are both positive. Then it follows from (3.1.32) that

$$\frac{y}{2} \leq y' \,.$$

We distinguish four cases. If $y, y' \leq 1$, then we have

$$|\kappa(-y) - \kappa(-y')| = \left| \frac{1-y}{1-e^{-iy}} - \frac{1-y'}{1-e^{-iy'}} \right|$$

and by the fundamental theorem of calculus

$$= \left| \int_{y'}^{y} \frac{-1 + e^{-it} + i(1-t)e^{it}}{(1-e^{-it})^2} \, dt \right| .$$

Using $y' \geq \frac{y}{2}$ and Theorem 3.1.9, we bound this by

$$\leq |y-y'| \sup_{\frac{y}{2} \leq t \leq 1} \frac{3}{|1-e^{-it}|^2} \leq 3|y-y'| (2\frac{2}{y})^2 \leq 2^6 \frac{|y-y'|}{|y|^2} \,.$$

If $y \leq 1$ and $y' > 1$, then $\kappa(-y') = 0$ and we have from the first case

$$|\kappa(-y) - \kappa(-y')| = |\kappa(-y) - \kappa(-1)| \leq 2^6 \frac{|y-1|}{|y|^2} \leq 2^6 \frac{|y-y'|}{|y|^2} \,.$$

Similarly, if $y > 1$ and $y' \leq 1$, then $\kappa(-y) = 0$ and we have from the first case

$$|\kappa(-y) - \kappa(-y')| = |\kappa(-y') - \kappa(-1)| \leq 2^6 \frac{|y'-1|}{|y'|^2} \leq 2^6 \frac{|y-y'|}{|y'|^2}\,.$$

Using again $y' \geq \frac{y}{2}$, we bound this by

$$\leq 2^6 \frac{|y-y'|}{|y/2|^2} = 2^8 \frac{|y-y'|}{|y|^2}$$

Finally, if $y, y' > 1$ then

$$|\kappa(-y) - \kappa(-y')| = 0 \leq 2^8 \frac{|y-y'|}{|y|^2}\,.$$

□

3.2 Smooth functions.

Lemma 3.2.1. *Let $f : \mathbb{R} \to \mathbb{C}$ be 2π-periodic and continuously differentiable, and let $n \in \mathbb{Z} \setminus \{0\}$. Then*

$$\hat{f}_n = \frac{1}{in}\widehat{f'}_n. \tag{3.2.1}$$

Proof. This is part of the Lean library. □

Lemma 3.2.2. *Let $f : \mathbb{R} \to \mathbb{C}$ such that*

$$\sum_{n\in\mathbb{Z}} |\hat{f}_n| < \infty. \tag{3.2.2}$$

Then

$$\sup_{x\in[0,2\pi]} |f(x) - S_N f(x)| \to 0 \tag{3.2.3}$$

as $N \to \infty$.

Proof. This is part of the Lean library. □

Lemma 3.2.3. *Let $f : \mathbb{R} \to \mathbb{C}$ be 2π-periodic and twice continuously differentiable. Then*

$$\sup_{x\in[0,2\pi]} |f(x) - S_N f(x)| \to 0 \tag{3.2.4}$$

as $N \to \infty$.

Proof. By Theorem 3.2.2, it suffices to show that the Fourier coefficients $\hat{f}_n$ are summable. Applying Theorem 3.2.1 twice and using the fact that f'' is continuous and thus bounded on $[0, 2\pi]$, we compute

$$\sum_{n\in\mathbb{Z}} |\hat{f}_n| = |\hat{f}_0| + \sum_{n\in\mathbb{Z}\setminus\{0\}} \frac{1}{n^2}|\widehat{f''}_n| \leq |\hat{f}_0| + \left(\sup_{x\in[0,2\pi]} |f(x)|\right) \cdot \sum_{n\in\mathbb{Z}\setminus\{0\}} \frac{1}{n^2} < \infty.$$

□

Proof of Theorem 3.1.2. Theorem 3.1.2 now follows directly from the previous Theorem 3.2.3. □

3.3 The truncated Hilbert transform

Let M_n be the modulation operator acting on measurable 2π-periodic functions defined by

$$M_n g(x) = g(x)e^{inx}\,. \tag{3.3.1}$$

Define the approximate Hilbert transform by

$$L_N g = \frac{1}{N}\sum_{n=0}^{N-1} M_{n+N} S_{N+n} M_{-N-n} g\,. \tag{3.3.2}$$

Lemma 3.3.1 (modulated averaged projection)**.** *We have for every bounded measurable 2π-periodic function g*

$$\|L_N g\|_{L^2[0,2\pi]} \le \|g\|_{L^2[0,2\pi]}\,. \tag{3.3.3}$$

Proof. We have

$$\|M_n g\|^2_{L^2[0,2\pi]} = \int_0^{2\pi} |e^{inx} g(x)|^2\,dx = \int_0^{2\pi} |g(x)|^2\,dx = \|g\|^2_{L^2[0,2\pi]}\,. \tag{3.3.4}$$

We have by the triangle inequality, the square root of the identity in (3.3.4), and Theorem 3.1.10

$$\begin{aligned}\|L_n g\|_{L^2[0,2\pi]} &= \|\frac{1}{N}\sum_{n=0}^{N-1} M_{n+N} S_{N+n} M_{-N-n} g\|_{L^2[0,2\pi]}\\ \le \frac{1}{N}\sum_{n=0}^{N-1} \|M_{n+N} S_{N+n} M_{-N-n} g\|_{L^2[0,2\pi]} &= \frac{1}{N}\sum_{n=0}^{N-1} \|S_{N+n} M_{-N-n} g\|_{L^2[0,2\pi]}\\ \le \frac{1}{N}\sum_{n=0}^{N-1} \|M_{-N-n} g\|_{L^2[0,2\pi]} &= \frac{1}{N}\sum_{n=0}^{N-1} \|g\|_{L^2[0,2\pi]} = \|g\|_{L^2[0,2\pi]}\,.\end{aligned} \tag{3.3.5}$$

This proves (3.3.3) and completes the proof of the lemma. □

Lemma 3.3.2 (periodic domain shift)**.** *Let f be a bounded 2π-periodic function. We have for any $0 \le x \le 2\pi$ that*

$$\int_0^{2\pi} f(y)\,dy = \int_{-x}^{2\pi-x} f(y)\,dy = \int_{-\pi}^{\pi} f(y-x)\,dy\,. \tag{3.3.6}$$

Proof. We have by periodicity and change of variables

$$\int_{-x}^{0} f(y)\,dy = \int_{-x}^{0} f(y+2\pi)\,dy = \int_{2\pi-x}^{2\pi} f(y)\,dy\,. \tag{3.3.7}$$

We then have by breaking up the domain of integration and using (3.3.7)

$$\int_0^{2\pi} f(y)\,dy = \int_0^{2\pi-x} f(y)\,dy + \int_{2\pi-x}^{2\pi} f(y)\,dy$$

$$= \int_0^{2\pi - x} f(y)\, dy + \int_{-x}^{0} f(y)\, dy = \int_{-x}^{2\pi - x} f(y)\, dy\,. \tag{3.3.8}$$

This proves the first identity of the lemma. The second identity follows by substitution of y by $y - x$. □

Lemma 3.3.3 (Young convolution)**.** *Let f and g be two bounded non-negative measurable 2π-periodic functions on $\mathbb{R}$. Then*

$$\left(\int_0^{2\pi} \left(\int_0^{2\pi} f(y) g(x-y)\, dy \right)^2 dx \right)^{\frac{1}{2}} \leq \|f\|_{L^2[0,2\pi]} \|g\|_{L^1[0,2\pi]}\,. \tag{3.3.9}$$

Proof. Using Fubini and Theorem 3.3.2, we observe

$$\int_0^{2\pi} \int_0^{2\pi} f(y)^2 g(x-y)\, dy\, dx = \int_0^{2\pi} f(y)^2 \int_0^{2\pi} g(x-y)\, dx\, dy$$

$$= \int_0^{2\pi} f(y)^2 \int_0^{2\pi} g(x)\, dx dy = \|f\|^2_{L^2[0,2\pi]} \|g\|_{L^1[0,2\pi]}\,. \tag{3.3.10}$$

Let h be the nonnegative square root of g, then h is bounded and 2π-periodic with $h^2 = g$. We estimate the square of the left-hand side of (3.3.9) with Cauchy-Schwarz and then with (3.3.10) by

$$\int_0^{2\pi} (\int_0^{2\pi} f(y) h(x-y) h(x-y)\, dy)^2\, dx$$

$$\leq \int_0^{2\pi} \left(\int_0^{2\pi} f(y)^2 g(x-y)\, dy \right) \left(\int_0^{2\pi} g(x-y)\, dy \right) dx$$

$$= \|f\|^2_{L^2[0,2\pi]} \|g\|^2_{L^1[0,2\pi]}\,.$$

Taking square roots, this proves the lemma. □

For $0 < r < 1$, Define the kernel k_r to be the 2π-periodic function

$$k_r(x) := \min\left(r^{-1}, 1 + \frac{r}{|1 - e^{ix}|^2} \right)\,, \tag{3.3.11}$$

where the minimum is understood to be r^{-1} in case $1 = e^{ix}$.

Lemma 3.3.4 (integrable bump convolution)**.** *Let g, f be bounded measurable 2π-periodic functions. Let $0 < r < \pi$. Assume we have for all x*

$$|g(x)| \leq k_r(x)\,. \tag{3.3.12}$$

Let

$$h(x) = \int_0^{2\pi} f(y) g(x-y)\, dy\,. \tag{3.3.13}$$

Then

$$\|h\|_{L^2[0,2\pi]} \leq 17 \|f\|_{L^2[-\pi,\pi]}\,. \tag{3.3.14}$$

Proof. From monotonicity of the integral and (3.3.12),

$$\|g\|_{L^1[0,2\pi]} \le \int_0^{2\pi} k_r(x)\,dx\,. \tag{3.3.15}$$

Using the symmetry $k_r(x) = k_r(-x)$, the assumption, and Theorem 3.1.9, the last display is equal to

$$\begin{aligned} &= 2\int_0^{\pi} \min\left(\frac{1}{r}, 1 + \frac{r}{|1-e^{ix}|^2}\right) dx \\ &\le 2\int_0^{r} \frac{1}{r}\,dx + 2\int_r^{\pi} 1 + \frac{64r}{x^2}\,dx \\ &\le 2 + 2\pi + 2\left(\frac{4r}{r} - \frac{4r}{\pi}\right) \le 17\,. \end{aligned} \tag{3.3.16}$$

Together with Theorem 3.3.3, this proves the lemma. □

Lemma 3.3.5 (Dirichlet approximation)**.** *Let $0 < r < 1$. Let N be the smallest integer larger than $\frac{1}{r}$. There is a 2π-periodic continuous function L' on $\mathbb{R}$ that satisfies for all $0 \le x \le 2\pi$ and all 2π-periodic bounded measurable functions f on $\mathbb{R}$*

$$L_N f(x) = \frac{1}{2\pi}\int_0^{2\pi} f(y)L'(x-y)\,dy\,. \tag{3.3.17}$$

Moreover, for all $-\pi \le x \le \pi$,

$$\left|L'(x) - \mathbf{1}_{\{y:\, r<|y|<1\}}\kappa(x)\right| \le 12k_r(x)\,. \tag{3.3.18}$$

Proof. We have by definition and Theorem 3.1.8

$$L_N g(x) = \frac{1}{N}\sum_{n=0}^{N-1}\int_0^{2\pi} e^{-i(N+n)x}K_{N+n}(x-y)e^{i(N+n)y}g(y)\,dy\,. \tag{3.3.19}$$

This is of the form (3.3.17) with the continuous function

$$L'(x) = \frac{1}{N}\sum_{n=0}^{N-1} K_{N+n}(x)e^{-i(N+n)x}\,. \tag{3.3.20}$$

With (3.1.22) of Theorem 3.1.8 we have $|K_N(x)| \le 2N+1$ for every x and thus

$$|L'(x)| \le \frac{1}{N}\sum_{n=0}^{N-1}(2N+2n+1) \le 4N \le 2^3 r^{-1}\,. \tag{3.3.21}$$

Therefore, for $|x| \in [0,r) \cup (1,\pi]$, we have

$$\left|L'(x) - \mathbf{1}_{\{y:\, r<|y|<1\}}(x)\kappa(x)\right| = |L'(x)| \le 2^3 r^{-1}. \tag{3.3.22}$$

This proves (3.3.18) for $|x| \in [0,r)$ since $k_r(x) = r^{-1}$ in this case.

For $e^{ix} \neq 1$ one may use the expression (3.1.23) for K_N in Theorem 3.1.8 to obtain

$$\begin{aligned} L'(x) &= \frac{1}{N}\sum_{n=0}^{N-1}\left(\frac{e^{i(N+n)x}}{1-e^{-ix}} + \frac{e^{-i(N+n)x}}{1-e^{ix}}\right)e^{i(N+n)x} \\ &= \frac{1}{N}\sum_{n=0}^{N-1}\left(\frac{e^{i2(N+n)x}}{1-e^{-ix}} + \frac{1}{1-e^{ix}}\right) \\ &= \frac{1}{1-e^{ix}} + \frac{1}{N}\frac{e^{i2Nx}}{1-e^{-ix}}\sum_{n=0}^{N-1} e^{i2nx} \end{aligned} \tag{3.3.23}$$

and thus

$$L'(x) - \mathbf{1}_{\{y:\, r<|y|<1\}}\kappa(x) = L''(x) + \frac{1-\mathbf{1}_{\{y:\, r<|y|<1\}}(x)(1-|x|)}{1-e^{ix}}, \tag{3.3.24}$$

where

$$L''(x) := \frac{1}{N}\frac{e^{i2Nx}}{1-e^{-ix}}\sum_{n=0}^{N-1} e^{i2nx}.$$

For $x \in [-\pi, r] \cup [r, \pi]$, we have using Theorem 3.1.9 that

$$\begin{aligned} \left|\frac{1-\mathbf{1}_{\{y:\, r<|y|<1\}}(x)(1-|x|)}{1-e^{ix}}\right| &= \left|\frac{\min(|x|,1)}{1-e^{ix}}\right| \leq \frac{2\min(|x|,1)}{|x|} \\ &\leq 2\cdot 1 \leq 2k_r(x). \end{aligned} \tag{3.3.25}$$

Next, we need to estimate $L''(x)$. If the real part of e^{ix} is negative, we have

$$1 \leq |1-e^{-ix}| \leq 2\,. \tag{3.3.26}$$

and hence

$$|L''(x)| \leq \frac{1}{N}\sum_{n=0}^{N-1} 1 = 1 \leq 1 + \frac{r}{|1-e^{ix}|^2}\,. \tag{3.3.27}$$

If the real part of e^{ix} is positive and in particular while still $e^{ix} \neq \pm 1$, then we have by telescoping

$$(1-e^{2ix})\sum_{n=0}^{N-1} e^{i2nx} = 1 - e^{i2Nx}\,. \tag{3.3.28}$$

As $e^{2ix} \neq 1$, we may divide by $1-e^{2ix}$ and insert this into (3.3.23) to obtain

$$L''(x) = \frac{1}{N}\frac{e^{i2Nx}}{1-e^{-ix}}\frac{1-e^{i2Nx}}{1-e^{2ix}}\,. \tag{3.3.29}$$

Hence, using the nonnegativity of the real part of e^{ix}

$$|L''(x)| \le \frac{2}{N}\frac{1}{|1-e^{ix}|}\frac{1}{|1-e^{2ix}|}$$

$$= \frac{2}{N}\frac{1}{|1-e^{ix}|^2}\frac{1}{|1+e^{ix}|} \le \frac{2r}{|1-e^{ix}|^2} \le 2\left(1+\frac{r}{|1-e^{ix}|^2}\right) \tag{3.3.30}$$

Moreover, for $|x| \in [r,\pi]$, one checks easily that

$$1+\frac{r}{|1-e^{ix}|^2} \le 5k_r(x).$$

Therefore, (3.3.27) and (3.3.30) show that, in this range of x,

$$|L''(x)| \le 10k_r(x).$$

Together with (3.3.25), this prove (3.3.18) for $|x| \in [r,\pi]$. As the range $|x| \in [0,r)$ is covered by (3.3.22), this completes the proof of the lemma. □

We now prove Theorem 3.1.6.

Proof of Theorem 3.1.6. We first show that if f is supported in $[1,4]$, then

$$\|H_r f\|_{L^2[2,3]} \le 2^8 \|f\|_{L^2(\mathbb{R})}\,. \tag{3.3.31}$$

Let $\tilde{f}$ be the 2π-periodic extension of f to $\mathbb{R}$. Let N be the smallest integer larger than $\frac{1}{r}$. Then, Equation (3.3.18) shows that the kernels of H_r and $2\pi L_N$ differ by at most $12k_r$ on $[-\pi,\pi]$. Consider $x \in [2,3]$. When computing $H_r f(x)$ and $2\pi L_N f(x)$, the kernels are computed at points of the form $x-y$ with $f(y) \neq 0$, i.e., $y \in [1,4]$. As $x \in [2,3]$, the difference $x-y$ is bounded in absolute value by 2, and therefore belongs to $[-\pi,\pi]$, where the above bound holds.

It follows that, for $x \in [2,3]$,

$$|H_r\tilde{f}(x)| \le 2\pi|L_N\tilde{f}(x)| + 12\left|\int_0^{2\pi} k_r(x-y)\tilde{f}(y)\,dy\right|.$$

Taking L^2 norm and using its sub-additivity, we get

$$\|H_r\tilde{f}\|_{L^2[2,3]} \le 2\pi\|L_N\tilde{f}\|_{L^2[0,2\pi]} + 12\left(\int_0^{2\pi}\left|\int_0^{2\pi} k_r(x-y)\tilde{f}(y)\,dy\right|^2 dx\right)^{\frac{1}{2}}.$$

Since κ is supported in $[-1,1]$, we have that $H_r\tilde{f}$ agrees with $H_r f$ on $[2,3]$. Using Theorem 3.3.1 and Theorem 3.3.4, we conclude

$$\|H_r f\|_{L^2[2,3]} \le 2\pi\|f\|_{L^2[0,2\pi]} + 12\cdot 17\|f\|_{L^2[0,2\pi]}\,, \tag{3.3.32}$$

which gives (3.3.31).

Suppose now that f is supported in $[c,c+3]$ for some $c \in \mathbb{R}$. Then the function $g(x) = f(x-c+1)$ is supported in $[1,4]$. By a change of variables in (3.1.18), we have $H_r g(x) = H_r f(x-c+1)$. Thus, by (3.3.31)

$$\|H_r f\|_{L^2[c+1,c+2]} = \|H_r g\|_{L^2[2,3]} \le 2^8\|g\|_2 = 2^8\|f\|_2\,. \tag{3.3.33}$$

Let now f be arbitrary. Since $\kappa(x) = 0$ for $|x| > 1$, we have for all $x \in [c+1, c+2]$

$$H_r f(x) = H_r(f\mathbf{1}_{[c,c+3]})(x)\,.$$

Thus

$$\int_{c+1}^{c+2} |H_r f(x)|^2\,\mathrm{d}x = \int_{c+1}^{c+2} |H_r(f\mathbf{1}_{[c,c+3]})(x)|^2\,\mathrm{d}x\,.$$

Applying the bound (3.3.33), this is

$$\leq 2^{16} \int_c^{c+3} |f(x)|^2\,\mathrm{d}x\,.$$

Summing over all $c \in \mathbb{Z}$, we obtain

$$\int_{\mathbb{R}} |H_r f(x)|^2\,\mathrm{d}x \leq 3 \cdot 2^{16} \int_{\mathbb{R}} |f(x)|^2\,\mathrm{d}x\,.$$

This completes the proof. □

3.4 The proof of the van der Corput Lemma

Proof of Theorem 3.1.7. Let g be a Lipschitz continuous function as in the lemma. Assume first that $n = 0$. Then

$$\int_\alpha^\beta g(x)\,\mathrm{d}x \leq |\beta - \alpha| \sup_{\alpha \leq x \leq \beta} |g(x)| \leq |\beta - \alpha| \|g\|_{Lip(\alpha,\beta)} (1 + |n||\beta - \alpha|)^{-1}$$

Assume now $n \neq 0$. Without loss of generality, we may assume $n > 0$. We distinguish two cases. If $\beta - \alpha < \frac{\pi}{n}$, we have by the triangle inequality

$$\left| \int_\alpha^\beta g(x) e^{inx}\,\mathrm{d}x \right| \leq |\beta - \alpha| \sup_{x \in [\alpha,\beta]} |g(x)| \leq 2\pi |\beta - \alpha| \|g\|_{Lip(\alpha,\beta)} (1 + |n||\beta - \alpha|)^{-1}\,.$$

We turn to the case $\frac{\pi}{n} \leq \beta - \alpha$. We have

$$e^{in(x+\pi/n)} = -e^{inx}\,.$$

Using this, we write

$$\int_\alpha^\beta g(x) e^{inx}\,\mathrm{d}x = \frac{1}{2} \int_\alpha^\beta g(x) e^{inx}\,\mathrm{d}x - \frac{1}{2} \int_\alpha^\beta g(x) e^{in(x+\pi/n)})\,\mathrm{d}x\,.$$

We split the first integral at $\alpha + \frac{\pi}{n}$ and the second one at $\beta - \frac{\pi}{n}$, and make a change of variables in the second part of the first integral to obtain

$$= \frac{1}{2} \int_\alpha^{\alpha + \frac{\pi}{n}} g(x) e^{inx}\,\mathrm{d}x - \frac{1}{2} \int_{\beta - \frac{\pi}{n}}^\beta g(x) e^{in(x+\pi/n)}\,\mathrm{d}x$$

$$+ \frac{1}{2} \int_{\alpha + \frac{\pi}{n}}^\beta (g(x) - g(x - \frac{\pi}{n})) e^{inx}\,\mathrm{d}x\,.$$

The sum of the first two terms is by the triangle inequality bounded by

$$\frac{\pi}{n} \sup_{x\in[\alpha,\beta]} |g(x)| \,.$$

The third term is by the triangle inequality at most

$$\begin{aligned} &\frac{1}{2}\int_{\alpha+\frac{\pi}{n}}^{\beta} |g(x) - g(x-\frac{\pi}{n})| \,\mathrm{d}x \\ \leq\ &\frac{|\beta-\alpha|}{2}\frac{\pi}{n} \sup_{\alpha\leq x<y\leq\beta} \frac{|g(x)-g(y)|}{|x-y|} \,. \end{aligned}$$

Adding the two terms, we obtain

$$\left|\int_{\alpha}^{\beta} g(x)e^{-inx} \,\mathrm{d}x\right| \leq \frac{\pi}{n}\|g\|_{\mathrm{Lip}(\alpha,\beta)} \,.$$

This completes the proof of the lemma, using that with $\frac{\pi}{n} \leq \beta - \alpha$,

$$\frac{\pi}{n} = \frac{2\pi|\beta-\alpha|}{2n|\beta-\alpha|} \leq 2\pi|\beta-\alpha|(1+n|\beta-\alpha|)^{-1} \,.$$

□

3.5 Partial sums as orthogonal projections

This subsection proves Theorem 3.1.10

Proof of Theorem 3.1.10. The functions $e_n : x \mapsto e^{inx}$ form an orthonormal basis in $L^2[-\pi,\pi]$ (this is already in Mathlib). Therefore we have

$$\begin{aligned} \|S_N f\|^2_{L^2[-\pi,\pi]} &= \|\sum_{n=-N}^{N} \langle \hat{f}_n, e_n\rangle_{L^2[-\pi,\pi]}\|^2_{L^2[-\pi,\pi]} \\ &= \sum_{n=-N}^{N} |\hat{f}_n| \\ &\leq \sum_{n\in\mathbb{Z}} |\hat{f}_n| \\ &= \|f\|_{L^2[-\pi,\pi]}. \end{aligned}$$

This completes the proof of the lemma. □

3.6 The error bound

Lemma 3.6.1 (Dirichlet kernel - Hilbert kernel relation)**.** *For all $N \in \mathbb{Z}$ and $x \in [-\pi, \pi] \setminus \{0\}$,*

$$\left|K_N(x) - (e^{-iNx}\kappa(x) + \overline{e^{-iNx}\kappa(x)})\right| \leq \pi\,.$$

Proof. Let $N \in \mathbb{Z}$ and $x \in [-\pi, \pi] \setminus \{0\}$. With Theorem 3.1.8, we obtain

$$K_N(x) - (e^{-iNx}\kappa(x) + \overline{e^{-iNx}\kappa(x)}) = e^{-iNx}\frac{\min(|x|,1)}{1-e^{ix}} + e^{iNx}\frac{\min(|x|,1)}{1-e^{-ix}}\,.$$

Using Theorem 3.1.9 with $\eta = \min(|x|, 1)$, we bound

$$\left|K_N(x) - (e^{-iNx}\kappa(x) + \overline{e^{-iNx}\kappa(x)})\right| \leq \frac{\min(|x|,1)}{|1-e^{ix}|} + \frac{\min(|x|,1)}{|1-e^{-ix}|} \leq \frac{\pi}{2} + \frac{\pi}{2} = \pi\,.$$

□

Lemma 3.6.2 (partial Fourier sum bound)**.** *Let $g : \mathbb{R} \to \mathbb{C}$ be a measurable 2π-periodic function such that for some $\delta > 0$ and every $x \in \mathbb{R}$,*

$$|g(x)| \leq \delta\,. \tag{3.6.1}$$

Then for every $x \in [0, 2\pi]$ and $N > 0$,

$$|S_N g(x)| \leq \frac{1}{2\pi}(Tg(x) + T\bar{g}(x)) + \pi\delta.$$

Proof. Let $x \in [0, 2\pi]$ and $N > 0$. We have with Theorem 3.1.8

$$|S_N g(x)| = \frac{1}{2\pi}\left|\int_0^{2\pi} g(y)K_N(x-y)\,dy\right|.$$

We use 2π-periodicity of g and K_N to shift the domain of integration to obtain

$$= \frac{1}{2\pi}\left|\int_{x-\pi}^{x+\pi} g(y)K_N(x-y)\,dy\right|.$$

Using the triangle inequality, we split this as

$$\leq \frac{1}{2\pi}\left|\int_{x-\pi}^{x+\pi} g(y)\left(K_N(x-y) - \max(|x-y|,0)K_N(x-y)\right)\,dy\right| \tag{3.6.2}$$

$$+\frac{1}{2\pi}\left|\int_{x-\pi}^{x+\pi} g(y)\max(|x-y|,0)K_N(x-y)\,dy\right|. \tag{3.6.3}$$

Note that all integrals are well defined, since K_N is by (3.1.22) bounded by $2N+1$. Using that

$$\max(|x-y|,0)K_N(x-y) = e^{-iN(x-y)}\kappa(x-y) + \overline{e^{-iN(x-y)}\kappa(x-y)}\,, \tag{3.6.4}$$

Theorem 3.6.1 and (3.6.5), we bound (3.6.2) by

$$\frac{1}{2\pi}\int_{x-\pi}^{x+\pi} |g(y)| \left|K_N(x-y) - e^{-iN(x-y)}\kappa(x-y) + \overline{e^{-iN(x-y)}\kappa(x-y)}\right| dy \leq \pi\delta\,.$$

By dominated convergence and since $\kappa(x-y) = 0$ for $|x-y| > 1$, (3.6.3) equals

$$\frac{1}{2\pi}\lim_{r\to 0^+}\left|\int_{r<|x-y|<1} g(y)\max(|x-y|,0)K_N(x-y)\,dy\right|.$$

We bound the limit by a supremum and rewrite using (3.6.4),

$$\leq \frac{1}{2\pi}\sup_{r>0}\left|\int_{r<|x-y|<1} g(y)\left(e^{-iN(x-y)}\kappa(x-y) + \overline{e^{-iN(x-y)}\kappa(x-y)}\right)\,dy\right|$$

Using the triangle inequality, we further bound this by

$$\begin{aligned}\leq\ & \frac{1}{2\pi}\sup_{r>0}\left|\int_{r<|x-y|<1} g(y)e^{-iNy}\kappa(x-y)\,dy\right| \\ & + \frac{1}{2\pi}\sup_{r>0}\left|\int_{r<|x-y|<1} \overline{g}(y)e^{-iNy}\kappa(x-y)\,dy\right|.\end{aligned}$$

By the definition (3.1.16) of T, this is

$$\leq \frac{1}{2\pi}(Tg(x) + T\overline{g}(x))\,.$$

□

Lemma 3.6.3 (real Carleson operator measurable)**.** *Let f be a bounded measurable function on $\mathbb{R}$. Then Tf as defined in* (3.1.16) *is measurable.*

Proof. Since a countable supremum of measurable functions is measurable, it suffices to show that for every $n \in \mathbb{Z}$,

$$x \mapsto \sup_{r>0}\left|\int_{r<|x-y|<1} f(y)\kappa(x-y)e^{iny}\,dy\right|$$

is measurable. So let $n \in \mathbb{Z}$. Note that for each $x \in \mathbb{R}$, the function

$$r \mapsto \left|\int_{r<|x-y|<1} f(y)\kappa(x-y)e^{iny}\,dy\right|$$

is continuous on $(0,\infty)$ since the integrand is locally bounded on the domain $0 < |x-y| < 1$ by the assumptions on f and Theorem 3.1.11. Thus, for each $x \in \mathbb{R}$,

$$\sup_{r>0}\left|\int_{r<|x-y|<1} f(y)\kappa(x-y)e^{iny}\,dy\right| = \sup_{r\in\mathbb{Q}_{>0}}\left|\int_{r<|x-y|<1} f(y)\kappa(x-y)e^{iny}\,dy\right|$$

The right hand side is again a countable supremum so it remains to show that for every $r \in \mathbb{Q}_{>0}$,

$$x \mapsto \left| \int_{r<|x-y|<1} f(y)\kappa(x-y)e^{iny}\,dy \right| = \left| \int \mathbf{1}_{\{r<|x-\cdot|<1\}}(y) f(y)\kappa(x-y)e^{iny}\,dy \right|$$

is measurable, which follows from the fact that the integrand is measurable in (x, y). □

Lemma 3.6.4 (partial Fourier sums of small)**.** *Let $g : \mathbb{R} \to \mathbb{C}$ be a measurable 2π-periodic function such that for some $\delta > 0$ and every $x \in \mathbb{R}$,*

$$|g(x)| \leq \delta\,. \tag{3.6.5}$$

Then for every $\epsilon > 0$, there exists a measurable set $E \subset [0, 2\pi]$ with $|E| < \epsilon$ such that for every $x \in [0, 2\pi] \setminus E$ and $N > 0$,

$$|S_N g(x)| \leq C_\epsilon \delta, \tag{3.6.6}$$

where

$$C_\epsilon = \left(\frac{8}{\pi\epsilon}\right)^{\frac{1}{2}} C_{4,2} + \pi\,. \tag{3.6.7}$$

Proof. Define

$$E := \{x \in [0, 2\pi] \ : \ \sup_{N>0} |S_N g(x)| > C_\epsilon \delta\}\,.$$

Then (3.6.6) clearly holds, and it remains to show that $|E| \leq \epsilon$. With Theorem 3.6.2, we obtain

$$E \subset \{x \in [0, 2\pi] \ : \ C_\epsilon \delta < \frac{1}{2\pi}(Tg(x) + T\bar{g}(x)) + \pi\delta\} \subset E_1 \cup E_2,$$

where

$$\begin{aligned} E_1 :=& \{x \in [0, 2\pi] \ : \ \pi(C_\epsilon - \pi)\delta < Tg(x)\} \\ E_2 :=& \{x \in [0, 2\pi] \ : \ \pi(C_\epsilon - \pi)\delta < T\bar{g}(x)\}. \end{aligned}$$

By Theorem 3.6.3, E_1 and E_2 are measurable. Thus,

$$\pi(C_\epsilon - \pi)\delta|E_1| \leq \int_{E_1} Tg(x)\,dx = \delta \int_{E_1} T(\delta^{-1} g \mathbf{1}_{[-\pi,3\pi]})(x)\,dx\,.$$

Applying Theorem 3.1.5 with $F = [-\pi, 3\pi]$ and $G = E'$, it follows that this is

$$\leq \delta \cdot C_{4,2}|F|^{\frac{1}{2}}|E_1|^{\frac{1}{2}} \leq (4\pi)^{\frac{1}{2}} C_{4,2}\delta \cdot |E'|^{\frac{1}{2}}\,.$$

Rearranging, we obtain

$$|E_1| \leq \left(\frac{(4\pi)^{\frac{1}{2}} C_{4,2}}{\pi(C_\epsilon - \pi)}\right)^2 = \frac{\epsilon}{2}\,.$$

Analogously, we get the same estimate for $|E_2|$. This completes the proof using $|E| \leq |E_1| + |E_2|$. □

Proof of Theorem 3.1.3. Theorem 3.1.3 follows directly from Theorem 3.6.4 with $\delta := \epsilon'$. □

3.7 Carleson on the real line

We prove Theorem 3.1.5.

Consider the standard distance function

$$\rho(x,y) = |x-y| \tag{3.7.1}$$

on the real line $\mathbb{R}$.

Lemma 3.7.1 (real line metric)**.** *The space $(\mathbb{R},\rho)$ is a complete locally compact metric space.*

Proof. This is part of the Lean library. □

Lemma 3.7.2 (real line ball)**.** *For $x \in R$ and $R > 0$, the ball $B(x,R)$ is the interval $(x-R, x+R)$*

Proof. Let $x' \in B(x,R)$. By definition of the ball, $|x'-x| < R$. It follows that $x'-x < R$ and $x-x' < R$. It follows $x' < x+R$ and $x' > x-R$. This implies $x' \in (x-R, x+R)$. Conversely, let $x' \in (x-R, x+R)$. Then $x' < x+R$ and $x' > x-R$. It follows that $x'-x < R$ and $x-x' < R$. It follows that $|x'-x| < R$, hence $x' \in B(x,R)$. This proves the lemma. □

We consider the Lebesgue measure μ on $\mathbb{R}$.

Lemma 3.7.3 (real line measure)**.** *The measure μ is a sigma-finite non-zero Radon-Borel measure on $\mathbb{R}$.*

Proof. This is part of the Lean library. □

Lemma 3.7.4 (real line ball measure)**.** *We have for every $x \in \mathbb{R}$ and $R > 0$*

$$\mu(B(x,R)) = 2R\,. \tag{3.7.2}$$

Proof. We have with Theorem 3.7.2

$$\mu(B(x,R)) = \mu((x-R, x+R)) = 2R\,. \tag{3.7.3}$$

□

Lemma 3.7.5 (real line doubling)**.** *We have for every $x \in \mathbb{R}$ and $R > 0$*

$$\mu(B(x,2R)) = 2\mu(B(x,R))\,. \tag{3.7.4}$$

Proof. We have with Theorem 3.7.4

$$\mu(B(x,2R) = 4R = 2\mu(B(x,R))\,. \tag{3.7.5}$$

This proves the lemma. □

The preceding four lemmas show that $(\mathbb{R},\rho,\mu,4)$ is a doubling metric measure space. Indeed, we even show that $(\mathbb{R},\rho,\mu,1)$ is a doubling metric measure space, but we may relax the estimate in Theorem 3.7.5 to conclude that $(\mathbb{R},\rho,\mu,4)$ is a doubling metric measure space.

For each $n \in \mathbb{Z}$ define $\vartheta_n : \mathbb{R} \to \mathbb{R}$ by

$$\vartheta_n(x) = nx\,. \tag{3.7.6}$$

Let Θ be the collection $\{\vartheta_n, n \in \mathbb{Z}\}$. Note that for every $n \in \mathbb{Z}$ we have $\vartheta_n(0) = 0$. Define

$$d_{B(x,R)}(\vartheta_n, \vartheta_m) := 2R|n-m|\,. \tag{3.7.7}$$

Lemma 3.7.6 (frequency metric)**.** *For every $R > 0$ and $x \in X$, the function $d_{B(x,R)}$ is a metric on Θ.*

Proof. This follows immediately from the fact that the standard metric on $\mathbb{Z}$ is a metric. □

Lemma 3.7.7 (oscillation control)**.** *For every $R > 0$ and $x \in X$, and for all $n, m \in \mathbb{Z}$, we have*

$$\sup_{y,y' \in B(x,R)} |ny - ny' - my + my'| \le 2|n-m|R\,. \tag{3.7.8}$$

Proof. The right hand side of (3.7.8) equals

$$\sup_{y,y' \in B(x,R)} |(n-m)(y-x) - (n-m)(y'-x)|\,.$$

The lemma then follows from the triangle inequality. □

Lemma 3.7.8 (frequency monotone)**.** *For any $x, x' \in X$ and $R, R' > 0$ with $B(x,R) \subset B(x,R')$, and for any $n, m \in \mathbb{Z}$*

$$d_{B(x,R)}(\vartheta_n, \vartheta_m) \le d_{B(x',R')}(\vartheta_n, \vartheta_m)\,.$$

Proof. This follows immediately from the definition (3.7.7) and $R \le R'$. □

Lemma 3.7.9 (frequency ball doubling)**.** *For any $x, x' \in \mathbb{R}$ and $R > 0$ with $x \in B(x', 2R)$ and any $n, m \in \mathbb{Z}$, we have*

$$d_{B(x',2R)}(\vartheta_n, \vartheta_m) \le 2 d_{B(x,R)}(\vartheta_n, \vartheta_m)\,. \tag{3.7.9}$$

Proof. With (3.7.7), both sides of (3.7.9) are equal to $4R|n-m|$. This proves the lemma. □

Lemma 3.7.10 (frequency ball growth)**.** *For any $x, x' \in \mathbb{R}$ and $R > 0$ with $B(x,R) \subset B(x', 2R)$ and any $n, m \in \mathbb{Z}$, we have*

$$2 d_{B(x,R)}(\vartheta_n, \vartheta_m) \le d_{B(x',2R)}(\vartheta_n, \vartheta_m)\,. \tag{3.7.10}$$

Proof. With (3.7.7), both sides of (3.7.9) are equal to $4R|n-m|$. This proves the lemma. □

Lemma 3.7.11 (integer ball cover)**.** *For every $x \in \mathbb{R}$ and $R > 0$ and every $n \in \mathbb{Z}$ and $R' > 0$, there exist $m_1, m_2, m_3 \in \mathbb{Z}$ such that*

$$B' \subset B_1 \cup B_2 \cup B_3\,, \tag{3.7.11}$$

where

$$B' = \{\vartheta \in \Theta : d_{B(x,R)}(\vartheta, \vartheta_n) < 2R'\} \tag{3.7.12}$$

and for $j = 1, 2, 3$

$$B_j = \{\vartheta \in \Theta : d_{B(x,R)}(\vartheta, \vartheta_{m_j}) < R'\}\,. \tag{3.7.13}$$

Proof. Let m_1 be the largest integer smaller than or equal to $n - \frac{R'}{2R}$. Let $m_2 = n$. Let m_3 be the smallest integer larger than or equal to $n + \frac{R'}{2R}$.

Let $\vartheta_{n'} \in B'$, then with (3.7.7), we have

$$2R|n - n'| < 2R' \,. \tag{3.7.14}$$

Assume first $n' \le m_1$. With (3.7.14) we have

$$\begin{aligned} R|m_1 - n'| &= R(m_1 - n') = R(m_1 - n) + R(n - n') \\ &< -\frac{R'}{2} + R' = -\frac{R'}{2} \,. \end{aligned} \tag{3.7.15}$$

We conclude $\vartheta_{n'} \in B_1$.

Assume next $m_1 < n' < m_3$. Then $\vartheta_{n'} \in B_2$.

Assume finally that $m_3 \le n'$. With (3.7.14) we have

$$\begin{aligned} R|m_3 - n'| &= R(n' - m_3) = R(n' - n) + R(n - m_3) \\ &< R' - \frac{R'}{2} = -\frac{R'}{2} \,. \end{aligned} \tag{3.7.16}$$

We conclude $\vartheta_{n'} \in B_1$. This completes the proof of the lemma. □

Lemma 3.7.12 (real van der Corput)**.** *For any $x \in \mathbb{R}$ and $R > 0$ and any function $\varphi : X \to \mathbb{C}$ supported on $B' = B(x, R)$ such that*

$$\|\varphi\|_{\mathrm{Lip}(B')} = \sup_{x \in B'} |\varphi(x)| + R \sup_{x,y \in B', x \neq y} \frac{|\varphi(x) - \varphi(y)|}{\rho(x, y)} \tag{3.7.17}$$

is finite and for any $n, m \in \mathbb{Z}$, we have

$$\left| \int_{B'} e(\vartheta_n(x) - \vartheta_m(x)) \varphi(x) d\mu(x) \right| \le 2\pi\mu(B') \frac{\|\varphi\|_{\mathrm{Lip}(B')}}{1 + d_{B'}(\vartheta_n, \vartheta_m)} \,. \tag{3.7.18}$$

Proof. Set $n' = n - m$. Then we have to prove

$$\left| \int_{x-R}^{x+R} e^{in'y} \varphi(y) dy \right| \le 4\pi R \|\varphi\|_{\mathrm{Lip}(B')} (1 + 2R|n'|)^{-1} \,. \tag{3.7.19}$$

This follows from Theorem 3.1.7 with $\alpha = x - R$ and $\beta = x + R$. □

Proof of Theorem 3.1.5. The preceding chain of lemmas establishes that Θ is a cancellative, compatible collection of functions on $(\mathbb{R}, \rho, \mu, 4)$. Again, some of the statements in these lemmas are stronger than what is needed for $a = 4$, but can be relaxed to give the desired conclusion for $a = 4$.

With κ as near (3.1.13), define the function $K : \mathbb{R} \times \mathbb{R} \to \mathbb{C}$ as in Theorem 1.1.1 by

$$K(x, y) := \kappa(x - y) \,. \tag{3.7.20}$$

The function K is continuous outside the diagonal $x = y$ and vanishes on the diagonal. Hence it is measurable.

By Theorems 3.1.11 and 3.1.12, it follows that K is a two-sided Calderón–Zygmund kernel on $(\mathbb{R}, \rho, \mu, 4)$. Theorem 3.1.6 verifies (2.0.3). Thus the assumptions of Theorem 2.0.1 are all satisfied. Applying the Theorem, Theorem 3.1.5 follows.

□

Theorem 3.7.13 (classical Carleson)**.** *Let f be a 2π-periodic complex-valued continuous function on $\mathbb{R}$. Then for almost all $x \in \mathbb{R}$ we have*

$$\lim_{N\to\infty} S_N f(x) = f(x), \tag{3.7.21}$$

where $S_N f$ is the N-th partial Fourier sum of f defined in (1.0.1) *with coefficients* (1.0.2)*.*

Proof of Theorem 3.7.13. By applying Theorem 3.1.4 with a sequence of $\epsilon_n := 2^{-n}\delta$ for $n \geq 1$ and taking the union of corresponding exceptional sets E_n, we see that outside a set of measure δ, the partial Fourier sums converge pointwise for $N \to \infty$. Applying this with a sequence of δ shrinking to zero and taking the intersection of the corresponding exceptional sets, which has measure zero, we see that the Fourier series converges outside a set of measure zero. □